Balachandar M

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Professional Summary

Module Lead and **Backend Engineer** with 3.5+ years of experience owning end-to-end delivery of complex backend and actuarial systems in Python and Go. **Led a team of 5** engineers, **reduced critical defects by 60%,** and drove three major platform migrations. Combines deep technical expertise in **microservices**, Redis, and JWT-based authentication with **hands-on leadership** — consistently delivering sprint commitments and translating dense technical requirements into measurable business outcomes.

Work Experience

Module Lead July 2024 - Present

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- **Led and mentored a team of 5 engineers** in designing and delivering high-fidelity actuarial computation systems; established a culture of precision that increased code quality by **25%** through structured reviews and enforced domain-specific validation standards.
- **Owned end-to-end project planning and sprint governance** for the actuarial module, decomposing dense mathematical specs into testable stories; reduced requirement ambiguity and shortened feedback cycles by **40%**.
- **Maintained zero team escalations** across **three** delivery cycles by implementing early-warning risk frameworks and pre-production validation gates, preventing an estimated **60%** of critical calculation defects from reaching live environments.
- **Coached and upskilled engineers** on dissecting complex actuarial logic and performance-tuning; improved team autonomy by **35%** and reduced senior-level intervention.
- **Architected and executed 3 major platform migrations** from legacy frameworks to modern architectures; delivered **100% data integrity** and zero downtime while achieving a **2x improvement in calculation speed** through optimized algorithm design.

Systems Engineer Nov 2023 - July 2024

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- **Partnered with cross-functional teams**—including Business Analysts (BAs) and Quality Assurance (QA), and domain experts—to translate ambiguous insurance requirements into technical specs; increased development velocity by **30%** and reduced mid-sprint requirement changes by **60%.**
- **Drove measurable improvements in system performance** by executing targeted refactoring; eliminated **30%** of dead code paths and reduced cyclomatic complexity by **25%,** resulting in a **40%** reduction in average response times for high-traffic modules.
- **Expanded technical ownership** by taking full accountability for **5** auxiliary modules—including valuation and charges layers—increasing the team's total delivery capacity by **30%** and eliminating critical single-point-of-failure risks.
- **Identified and resolved over 20+ latent logical defects** and edge-case failures across the application stack; elevated system uptime to **99.9%** and achieved a **90%** increase in positive client satisfaction scores regarding output quality.

Assistant Systems Engineer Aug 2022 - Nov 2023

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- **Engineered intricate business logic** for insurance policy projections; optimized the calculation engine to process high-volume of policy data, reducing total execution time from **20 minutes to 2–10 minutes** (a **50%** performance gain).
- **Independently drove the complete lifecycle** of the Quotation module; delivered the architecture and data modelling **weeks ahead of schedule** while ensuring **100% accuracy** against dense actuarial mathematical benchmarks.
- **Strengthened system reliability** by instituting testing protocols and clear debugging workflows; achieved **85% code coverage** and reduced post-deployment "calculation drift" or defects by **45%** compared to legacy baselines.
- **Accelerated release velocity** by hitting **100%** of Agile sprint commitments; maintained **100% on-time delivery** across **20+** consecutive sprints while rapidly resolving technical blockers for the team.

Skills

Python | Go | Actuarial Systems | Microservices | System Design | Django | Django REST Framework | FastAPI | Flask | Quart | Gin | PostgreSQL | REST API | GraphQL | WebSockets | Redis | RabbitMQ | Celery | Docker | Git | JWT Authentication | API Security | Performance Optimization | Caching | Database Indexing | Insurance Domain | Technical Business Analytics | CI/CD | Ubuntu Linux | Cloud Deployment | Linux/VPS Deployment | Agile

Projects

ConsTell Feb 2026 – Apr 2026 | Website Link | **Demo: SampleUser / Sample@211**

A **production-deployed**, 9-service **microservices** chat platform hosted on an **Ubuntu Server**, secured with **SSL/TLS**, custom domain, and **Nginx** as reverse proxy. Each service owns an **isolated PostgreSQL** DB, utilizing **Docker** for full-stack containerization and **Docker Compose** for automated CI/CD deployment pipelines.

- **API Gateway** (Source Code) - Go(Gin) **reverse-proxy** gateway centralizing routing and **security** for all 7 backend services. Enforces global middleware for **JWT validation**, **CORS enforcement**, and Redis-backed rate limiting.
- **Auth Service** (Source Code) - FastAPI RESTful service managing the full user lifecycle—registration, deletion, and forgot-password flows—with **JWT-based stateless auth**, **Pydantic** validation, and **Aerich** for schema migrations.
- **Profile Service** (Source Code) - Flask/GraphQL profile service using **Tortoise ORM** for user data retrieval and profile picture management. Uses **Alembic** for schema versioning and migration management.
- **Chat Service** (Source Code) - Go(Gin) service using **WebSockets** and **Redis Pub/Sub** for real-time bidirectional messaging. Provides **DEK**-based **end-to-end encryption** for both DMs and group rooms.
- **Chat DB Service** (Source Code) - Dedicated **PostgreSQL**-backed service handling message persistence, room configurations, participant metadata, and encryption keys—ensuring structured, **secure access** to historical chat data.
- **Nebula Bot** (Source Code) - AI assistant powered by **Llama-3.1-8b-instant** via Groq, delivering **context-aware**, real-time support through a natural language interface with a multi-model fallback chain for resilience.
- **Message Broker** (Source Code) - **RabbitMQ**-driven async task processor handling email triggers, cleanup routines, and message chunking workflows to maintain consistent state across all services.
- **Mailer API** (Source Code) - Async email service built with **Quart**, processing mail requests via background queues for non-blocking, **high-throughput** delivery.
- **Frontend** (Source Code) - Responsive **React** (Vite) featuring real-time chat, profile management, and a deep-space design system—acting as the unified orchestration layer across all backend services.

Games-Hub Oct 2025 - Dec 2025 Source Code Website Link

- Achieved **20x Performance Gain**: Engineered a high-performance backend architecture, utilizing Redis caching (per-view & object-level) and optimized PostgreSQL indexing, drastically reducing API latency for concurrent users.
- Fortified Security & Logging: Built a production-grade JWT authentication system with **Role-Based Access Control (RBAC)** and integrated a comprehensive user activity logging module, ensuring data integrity and protection against unauthorized access.
- Resilient Error Handling: Implemented a global exception handling strategy that captures and resolves 99% of runtime errors, ensuring high system availability and a seamless user experience
- Scalable Async Architecture: Offloaded resource-intensive tasks (emails, data processing) to background workers using **Celery** and **Redis**, preventing main-thread blocking and enhancing overall system responsiveness.
- DevOps & Deployment: Containerized the entire application ecosystem using **Docker** for consistent environments and deployed via Render, ensuring zero-downtime releases and easy scalability.
- Standardized API Ecosystem: Designed secure, well-documented REST APIs using Django Rest Framework and DRF Spectacular (**OpenAPI v3**), streamlining frontend integration and maintenance.

Awards

Star of The Quarter • Star of The Month • On The Spot Award • On The Spot Team Award • Certificate of Recognition

Education

St Peter's College of Engineering and Technology July 2018 – July 2022
Bachelors of Engineering Electronics and Communication Engineering, CGPA : **8.89**